

```
sudh@sudh-HP-Laptop-15s-fq5xxx:~/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user
/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm
user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_log
stress user/_forphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find
user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953
total 2000
balloc: first 1113 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -
nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=
raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ echo "Sudheer Rathore"
exec echo failed
$ echo "Sudheer Rathore"
"Sudheer Rathore"
$ echo "ERP: 31683"
"ERP: 31683"
$ echo "Problem Set #1 - Q1: sleep"
"Problem Set #1 - Q1: sleep"
$ sleep 2
$ sleep 20
$ █
```

```
$ echo
$ echo Sudheer_Rathore
Sudheer_Rathore
$ echo ERP_31683
ERP_31683
$ echo ProblemSet1_Q2_sixfive
ProblemSet1_Q2_sixfive
$ sixfive sixfive.txt
5
100
18
6
$ sixfive doesnotexist
sixfive: cannot open doesnotexist
$ sixfive
1 2 3 4 5 6 7 8 9 12 14 155 165
6
12
155
$
```

```
$ echo
$ echo Sudheer_Rathore
Sudheer_Rathore
$ echo ERP_31683
ERP_31683
$ echo ProblemSet1_Q3_memdump
ProblemSet1_Q3_memdump
$ memdump
Example 1:
61810
2026
Example 2:
a string
Example 3:
another
Example 4:
000000000000000d58
1819438967
100
z
xyzzzy
Example 5:
hello
w
o
r
l
d
$ echo a | memdump i
memdump: not enough data for 'i'
$
```

```
$  
$  
$ echo Sudheer_Rathore  
Sudheer_Rathore  
$ echo ERP_31683  
ERP_31683  
$ echo Problemset1_Q4_find  
Problemset1_Q4_find  
$ find . wc  
./wc  
$ echo > b  
$ mkdir a  
$ echo >a/b  
$ mkdir a/aa  
$ echo > a/aa/b  
$ find . b  
./b  
./a/b  
./a/aa/b  
$ █
```

```
$ echo
$ echo Sudheer_Rathore
Sudheer_Rathore
$ echo ERP_31683
ERP_31683
$ echo Problemset1_Q5_exec
Problemset1_Q5_exec
$ find . wc -exec echo FOUND
FOUND ./wc
$ find . wc -exec echo hi
hi ./wc
$ sh < findtest.sh
$ mkdir: a failed to create
$ $ $ $ $ hello
hello
hello
$ $
$
$
$
```

```
sudh@sudh-HP-Laptop-15s-fq5xxx:~/xv6-labs-2026$ echo "Sudheer Rathore 31683"
Sudheer Rathore 31683
sudh@sudh-HP-Laptop-15s-fq5xxx:~/xv6-labs-2026$ git status
On branch problemSet1
Your branch is up to date with 'origin/problemSet1'.

nothing to commit, working tree clean
sudh@sudh-HP-Laptop-15s-fq5xxx:~/xv6-labs-2026$ cat time.txt
7
sudh@sudh-HP-Laptop-15s-fq5xxx:~/xv6-labs-2026$ █
```

```
sudh@sudh-HP-Laptop-15s-fq5xxx:~/xv6-labs-2026$ git status
On branch problemSet1
Your branch is up to date with 'origin/problemSet1'.

nothing to commit, working tree clean
sudh@sudh-HP-Laptop-15s-fq5xxx:~/xv6-labs-2026$ echo "Sudheer Rathore 31683"
Sudheer Rathore 31683
sudh@sudh-HP-Laptop-15s-fq5xxx:~/xv6-labs-2026$ ./grade-lab-util
make: 'kernel/kernel' is up to date.
== Test sleep, no arguments == sleep, no arguments: OK (0.8s)
== Test sleep, returns == sleep, returns: OK (0.7s)
== Test sleep, makes syscall == sleep, makes syscall: OK (1.1s)
== Test sixfive_test == sixfive_test: OK (0.9s)
== Test sixfive_readme == sixfive_readme: OK (1.2s)
== Test sixfive_all == sixfive_all: OK (1.0s)
== Test memdump, examples == memdump, examples: OK (1.1s)
== Test memdump, format ii, S, p == memdump, format ii, S, p: OK (1.1s)
== Test find, in current directory == find, in current directory: OK (0.8s)
== Test find, in sub-directory == find, in sub-directory: OK (1.0s)
== Test find, recursive == find, recursive: OK (1.1s)
== Test exec == exec: OK (0.9s)
== Test exec, multiple args == exec, multiple args: OK (1.2s)
== Test exec via sh script == exec via sh script: OK (0.9s)
== Test sleep, zero and non-numeric args return == sleep, zero and non-numeric args
return: OK (0.9s)
== Test sixfive, stdin and missing file == sixfive, stdin and missing file: OK (1.1
s)
== Test memdump, formats h c, bounds checking, usage == memdump, formats h c, bound
s checking, usage: OK (0.8s)
== Test find, missing target, no duplicates, directory match == find, missing targe
t, no duplicates, directory match: OK (1.3s)
== Test time ==
time: OK
Score: 156/156
sudh@sudh-HP-Laptop-15s-fq5xxx:~/xv6-labs-2026$
```